Technical Interview Prep

Tooling Engineering
- BL0 Project
- PCU Holder Project

Process Engineering
- Capital Equipment
- MRSI Project
- Goldwing Program

Manufacturing Engineering
- Following a Product

To Dos: Engines

🔹 **How to Prep Smart**
- Review **core fundamentals** (not obscure equations).
- Be ready to **talk through your thinking**, not just answers.
- Prepare **2–3 strong project stories** you can reuse.
- Know the company's products and how mechanical engineering applies to them.

To Dos:
Establish Prompts
Answer:
Behavioral
Situational
Technical
Story Arsenal
Flash Cards to Rehearse & Practice
Rotary Machinery & Notes

📚 **Story Arsenal (ready to pull)**
Table: Type // Common Phrases Buzzwords // Story
Mentally confirm I have:
• ☐ One signature technical win // BL0
• ☐ One failure or recovery // MRSI
• ☐ One tradeoff decision // Bl0
• ☐ One cross-functional conflict // Goldwing
• ☐ One process improvement // Capital Project
• ☐ One leadership / ownership moment // Capital Project

Type	Common Phrase	Story

Technical win	Describe a challenging engineering problem you worked on.	**BL0 Project**
Failure or recovery	Tell me about a time you failed or made a mistake. What did you learn?	**MRSI Component**
Cross-functional conflict	Describe a time you worked on a team with conflicting opinions.	**Goldwing**
Tradeoff decision		**BL0 Project**
Process Improvement		**Capital Laser Weld**
Leadership / Ownership		**Capital Laser Weld**

Behavioral

Tell me about yourself.	[Insert Elevator Pitch]
Tell me about a time you failed or made a mistake. What did you learn?	Right is better than being rushed Delegating is better than trying to do it all Done is better than perfect
How do you handle tight deadlines or multiple projects?	
Give an example of when you had to explain a technical concept to a non-technical audience.	
Tell me about a time you took initiative or led a project.	
How do you handle feedback or criticism?	Understanding the bigger picture like the Why Giving a timeline or quantifiable metrics to measure past / future performance

Situational Questions (logic, ethics, communication, and practicality) "what would you do if…" questions to test judgment and decision-making.

What would you do if your design didn't meet safety	Acknowledge → Assess → Act → Communicate → Prevent Recurrence

requirements late in the project?	What they're testing: Ethics, safety mindset, courage under pressure. I'd assess the gap, propose corrective design changes, and evaluate the schedule and cost impact. I'd work with stakeholders to find the safest path forward rather than pushing something that could fail in the field. Example: Safety Project for the PCUs
How would you handle discovering an error in your calculations after submission?	**Integrity & Accountability** "I'd verify the error first, understand its impact, and then notify the appropriate people as soon as possible. I'd explain the mistake, provide corrected calculations, and outline whether it affects the design, safety, or timeline. I believe transparency builds trust, even when it's uncomfortable.
If a teammate isn't pulling their weight, how would you handle it?	**Team Skills & Conflict Management** "I'd start by having a private, respectful conversation to understand if there's a workload, skill, or communication issue. If needed, I'd offer help or clarify expectations. If the issue continued and affected the project, I'd bring it to the team lead with a focus on solutions, not blame." Example: Working with V on Goldwing
How would you prioritize tasks when multiple projects are due at the same time?	**Time Management & Judgement** "I'd prioritize based on deadlines, technical risk, and impact. I'd break tasks into smaller deliverables, identify critical paths, and communicate early if conflicts exist. If necessary, I'd align priorities with my manager so expectations are clear." Example: Tooling Projects // Process
What would you do if manufacturing tells you your design is too expensive to produce?	**DFM Mindset & Collaboration** I'd work with manufacturing to understand the cost drivers—materials, tolerances, processes—and look for design alternatives that meet performance requirements at lower cost. I see manufacturing feedback as part of optimizing the design, not a roadblock. Propose different solutions with the understanding of different risks
How would you respond if your manager disagreed	**Professionalism & Influence without ego** "I'd make sure I clearly explain my reasoning, data, and assumptions. If my manager still disagreed, I'd listen to their

with your technical recommendation?	perspective and understand their constraints. Ultimately, I'd support the final decision while documenting risks if appropriate." Clearly document the risks

Technical Questions

Bonus "Sneaky" Questions (These blend everything together):

Walk me through one of your projects from concept to completion.	
What's your favorite engineering project and why?	
What would you improve in this design? *(they may show you a part)*	
How do you ensure safety and compliance in your designs?	**Asking for feedback from different perspectives (Bridging the gap between Engineering & Technicians)**
Process Overview	FPA Laser Weld Empennage

Analysis & Tools

Have you used FEA or CFD? Explain your process.	

How do you validate simulation results?	
Experience with MATLAB, Python, or Excel?	**MATLAB – Hexapod System** **Python – UW College** **Excel – Data Export**
How do you approach troubleshooting a mechanical system?	

Materials

How do you select materials for a specific application?	
Difference between steel, aluminum, and cast iron?	
What is heat treatment?	
What is creep?	
What are composites and where are they used?	

Manufacturing & Design

Difference between machining, casting, forging, and additive manufacturing.	
What are GD&T symbols and why are they used?	
Explain tolerance and fit (clearance,	

interference, transition).	
What CAD tools have you used? (SolidWorks, CATIA, Inventor, etc.)	
How do you design for manufacturability (DFM)?	

Mechanical Fundamentals

What is Young's modulus?	
Difference between ductile and brittle materials?	
What causes fatigue failure?	
What is factor of safety and why is it important?	
Explain torsion in shafts.	

Thermo / Fluids (if relevant)

Difference between conduction, convection, and radiation.	
What is Reynolds number?	
Explain the first law of thermodynamics.	
Difference between laminar and turbulent flow?	
What is efficiency vs. effectiveness?	

Company Specific

What excites me about [Alten Technology]	
What makes me a good fit for [Alten Technology]	
Questions I have for [Alten]:	- Work Life Balance - Relationship with Clients

Insane Engineering of GenX Video Notes
Market Shares:
GE + CFM = 55%
Pratt & Whitney = 22%
Rolls Royce = 18%

Note: This video focuses on 787 Engine

- Cartridge Starts for starting the Turbine (Historical way to start the engine)
- Air Starting using Hover Cart or APU
- APU (Mini Turbine engine that uses electric battery that can start up Wing Engines)
    - APU Generator fires up Generator of Engine Motor to start up
- Hot Air typically needs to be exhausted out and can be inefficient
    - But 787 uses a specific system to transfer energy from hot compressed air to
- 787 uses Brakes actuated by Electric Motors instead of Hydraulic System Brakes
    - Reduced weight + easy to maintain & service + space

GenX Engines

- Each Engine provides 310 KN of Trust
- Previous Generation: CF6 (767)
- GenX uses 15% less fuel than previous generation
    - Reason 1: Bypass Duct & Bypass Ratio of 9:1

        - 9 kg Bypass Air to 1 kg Core Air
            - Previous was 5.7 Bypass Ratio
    - Downside is that the engines must be larger
    - Larger engines mean larger fan blades which can experience higher centrifugal force at the tips and deteriorate/damage the tips ($F=mw^2r$)
        - Reduce m using carbon fiber blades instead of titanium counterparts (strength to weight ratio)
        - Carbon Fiber Form Factor allows for 18 blades instead of 22 (reducing weight again)
    - Huge Downside of Composite: Impact Resistance: Chicken Gun Test
        - Composite Blades could not pass Chicken Gun Test
        - Front Interface of Blades now added with Titanium Reinforcements

To Be Continued: 16:23

Behavioral Questions (teamwork, communication, problem-solving)

Tell me about yourself.
Describe a challenging engineering problem you worked on.
Tell me about a time you failed or made a mistake. What did you learn?
Describe a time you worked on a team with conflicting opinions.
How do you handle tight deadlines or multiple projects?
Give an example of when you had to explain a technical concept to a non-technical audience.
Tell me about a time you took initiative or led a project.

Drafted responses:

- Tell me about yourself.
    - [Insert Elevator Pitch]
- Describe a challenging engineering problem you worked on.
    - [Tooling: BL0 Project]
    - [Materials & Process: Laser Weld Capital Project]
    - [Manufacturing: LVA-WHA Defects]
- Tell me about a time you failed or made a mistake. What did you learn?
    - []

    - Right is better than being rushed
    - Delegating is better than trying to do it all
    - Done is better than perfect
- Describe a time you worked on a team with conflicting opinions.
    - Ergonomics / Rate
- How do you handle tight deadlines or multiple projects?
    - Working backwards from the goal
    - Creating 1 big tasks into smaller tasks. Incorporating that into the schedule and see where our pinch points are
- Give an example of when you had to explain a technical concept to a non-technical audience.
    - Process & Tooling to Sports – Explaining what I do to Sports friends
- Tell me about a time you took initiative or led a project.
    - Capital Laser Weld Project
    - Draco Program from Start to Finish
    - Tooling PCU Project
    - Highlight understanding the process first and getting a better idea of it. Asking around for different opinions to understand different perspectives and consistent stories
- How do you handle feedback or criticism?
    - Understanding the bigger picture like the Why
    - Giving a timeline or quantifiable metrics to measure past / future performance

What would you do if your design didn't meet safety requirements late in the project?

How would you handle discovering an error in your calculations after submission?
If a teammate isn't pulling their weight, how would you handle it?
How would you prioritize tasks when multiple projects are due at the same time?
What would you do if manufacturing tells you your design is too expensive to produce?
How would you respond if your manager disagreed with your technical recommendation?

Drafted Responses:
Suggested Structure: Acknowledge à Assess à Act à Communicate à Prevent Recurrence

- What would you do if your design didn't meet safety requirements late in the project?
    - What they're testing: Ethics, safety mindset, courage under pressure.
    - I'd assess the gap, propose corrective design changes, and evaluate the schedule and cost impact. I'd work with stakeholders to find the safest path forward rather than pushing something that could fail in the field.
    - Example: Safety Project for the PCUs

How would you handle discovering an error in your calculations after submission?
Integrity & Accountability
"I'd verify the error first, understand its impact, and then notify the appropriate people as soon as possible. I'd explain the mistake, provide corrected calculations, and outline whether it affects the design, safety, or timeline. I believe transparency builds trust, even when it's uncomfortable.
Example?

If a teammate isn't pulling their weight, how would you handle it?

Team Skills & Conflict Management
"I'd start by having a private, respectful conversation to understand if there's a workload, skill, or communication issue. If needed, I'd offer help or clarify expectations. If the issue continued and affected the project, I'd bring it to the team lead with a focus on solutions, not blame."
Example: Working with V on Goldwing

How would you prioritize tasks when multiple projects are due at the same time?

Time Management & Judgement
"I'd prioritize based on deadlines, technical risk, and impact. I'd break tasks into smaller deliverables, identify critical paths, and communicate early if conflicts exist. If necessary, I'd align priorities with my manager so expectations are clear."
Example: Tooling Projects // Process

What would you do if manufacturing tells you your design is too expensive to produce?

DFM Mindset & Collaboration
I'd work with manufacturing to understand the cost drivers—materials, tolerances, processes—and look for design alternatives that meet performance requirements at lower cost. I see manufacturing feedback as part of optimizing the design, not a roadblock.
Propose different solutions with the understanding of different risks

How would you respond if your manager disagreed with your technical recommendation?

Professionalism & Influence without ego
"I'd make sure I clearly explain my reasoning, data, and assumptions. If my manager still disagreed, I'd listen to their perspective and understand their constraints. Ultimately, I'd support the final decision while documenting risks if appropriate."
Clearly document the risks

Technical Questions

- Focal Plane Array ELI5
- Laser Weld ELI5

- Airplane Manufacturing ELI5

🔹 **Bonus "Sneaky" Questions** (These blend everything together):

- Walk me through one of your projects from concept to completion.
- What's your favorite engineering project and why?
- What would you improve in this design? *(they may show you a part)*
- How do you ensure safety and compliance in your designs?

ALTEN PREP 2

✅ Pre-Interview Personal Checklist

🧠 Mindset (30 seconds)

☐ I'm here because I'm **qualified**, not because I got lucky
☐ This is a **technical conversation**, not an interrogation
☐ My goal is to show **how I think**, not to impress with volume

Quiet reframe: *They're deciding if they want to work with me.*

⏸️ Delivery Rules (remember these)

☐ Pause **before** answering
☐ Speak ~10% slower than normal
☐ Use silence instead of filler words
☐ 60–90 seconds, then stop

If I go long: **"Happy to go into more detail if needed."**

🧭 Answer Structure (every question)

☐ 1–2 sentence context
☐ State **my role** clearly
☐ Explain **reasoning / decisions**
☐ End with **result or lesson**

If stuck: "The key decision was…"

🎯 Content Focus

☐ Emphasize **judgment and tradeoffs**
☐ Say "I" when describing ownership
☐ Tie answers to **impact** (yield, risk, cost, schedule)
☐ Match technical depth to the interviewer

Avoid:

- ❌ Over-detail

- ❌ Apologizing
- ❌ Rambling backstory

📚 **Story Arsenal (ready to pull)**

Mentally confirm I have:
☐ One signature technical win
☐ One failure or recovery
☐ One tradeoff decision
☐ One cross-functional conflict
☐ One process improvement
☐ One leadership / ownership moment

I can reuse stories — that's okay.

👀 **Read & Adjust**
☐ Watch for nods / interruptions
☐ Pause after finishing — let them ask
☐ Ask for direction if unclear

Safe redirect: "Would you like more detail on the analysis or the outcome?"

🔚 **Strong Endings**
☐ Finish answers with a result or takeaway
☐ Stop talking once the point is made
☐ Let silence work for me

⭐ **Final Reminder (right before starting)**
☐ Be calm, curious, and professional
☐ I don't need to be perfect — I need to be **clear**
☐ Confidence = structure + pauses

Introductions

Thank you for taking the time to speaking with me today

I'm doing well, sun is out in Seattle, Seahawks are in the Superbowl, Life is good so far

Yes of course, very nice to meet you, thanks for having me today.

Outros

Thank you for taking the time to speaking with me today

I will look out for your next email.

Let me know if there are any further actions from my end

Tell Me More About Yourself

My background is in Mechanical Engineering – I hold both a Bachelors and Masters Degree in Mechanical Engineering.

I've been an engineer in the aerospace industry for about 10 years - with roles from Manufacturing to Materials to Process to Design Engineering.

So I've worn many hats throughout the industry and some of the products I've worked on include semiconductors, fighter jets, commercial aviation and space payloads.

I'm happy to go into detail for each of those roles/chapters but I would say a common theme across my career has been bridging engineering design and manufacturing execution to deliver critical flight hardware.

Tooling Engineer Overview

Context:
Currently I am a Tooling Design Engineer at Boeing for the 777 Program, specifically the empennage or the tail of the aircraft (Think Horizontal Stabilizer, Vertical Fin, Rudder, Tail Cone)

My Role:
My role is to design and develop tooling to support the factory with production and testing –

Some of that can be day to day tasks like tool fixes, dispositions, working with suppliers .... To.... more complex projects like when there's a design change or an update to the airplane configuration or addressing a safety incident.

Airplane Production

Context:
The Factory is broken up into control codes and I belong to a control code called MSDC (Major Structure Delivery Center). Our process is the beginning of the production line so we feed into the control codes of AFT Body Integration and eventually before Final Assembly.

My Role:
Because of this, I've learned that some of the decisions I make can make an impact later in the production line so I've been more intentional of being familiar with the downstream and upstream processes of the empennage even if I don't work with them consistently every day.

IDCA Production

Context:
So before Boeing, I worked on a product called Infrared Detector Cooler Assembly which is just a fancy term for a infrared sensor

But basically our products would be integrated into special locations of a fighter jet or space satellite

My Role:
And I supported this product as a Manufacturing Engineer and Process Engineer so I was able to see the product lifecycle with two similar but different lenses.

With one being more of a specialized role while another focused on the product lifecycle.

Manufacturing Engineering

Right out of college, I started out as a Manufacturing Engineer for a defense company (Raytheon Technologies) focused on building IDCAs — which is a fancy term for sensors that detect infrared. My team was responsible for our portfolio of Space Programs.

So my role was to learn the build plan and follow the hardware throughout the factory, make sure we were meeting engineering drawings and standard procedures, dispositioning defects, performing Root Cause Corrective Actions, First Article Inspections - basically taking the product from cradle to grave.

Reflection

It was honestly the best job to kick off my career because I was able to follow the product throughout the whole lifecycle and see so many different perspectives. I was able to work with technicians with decades of experience, and also work with customers and managers to see a high level view of a program.

It forced me to wear many hats—solving problems with different cross-functional teams, and diving into both the technical and practical sides of engineering. The diversity of challenges made me a true "jack of all trades."

Process Engineer Overview

I was in that Manufacturing role for a few years and I wanted to build my technical skills so I transitioned to a Materials & Process Engineering role.
And this role was exciting because, while Manufacturing is focusing on wearing multiple hats, Process Engineering is about wearing one hat really really well.

And so for example at Raytheon, one process that I was responsible for was Laser Welding Automation. And my job was essentially to become the Subject Matter Expert for that process.
And as different programs and derivatives of hardware went through Laser Weld Automation, it was my job to understand what requirements we needed to meet and realistically how consistent we can achieve these requirements (given the resources we have)

Reflection
This chapter helped me become more technical and step into the role of a Subject Matter Expert. It helped me understand the small details and nuances that make each part of the production cycle work smoothly. I learned that being a strong process engineer isn't just about understanding the theory; it's about applying it in real-world settings, balancing efficiency with quality, and constantly looking for ways to improve."

Strengths

"One of my key strengths is **my ability to see problems from multiple perspectives.** I think that's especially important in cross-functional environments, where different teams may have different priorities but are ultimately working toward the same objective.

A good example is the constant balance between cost, quality, and schedule. Each function naturally emphasizes a different part of that triangle, and I've found that understanding why each group has its priorities helps drive better decisions and smoother execution.

So **as someone who has worn many hats, I've often acted as a bridge between teams**, helping translate constraints and tradeoffs so we can move forward efficiently."

Weaknesses

“Earlier in my career, I tended to take on too much responsibility myself because I wanted to make sure things were done correctly.

Over time, I’ve learned that while ownership is important, leveraging the team and delegating effectively leads to better outcomes and stronger teams. I’m much more intentional now about aligning responsibilities early and trusting others’ expertise.”

Leaving Each Roles

"Each transition in my career has been driven by growth and development. Early on, I wanted to explore different parts of the engineering lifecycle to build a strong foundation.

For example, I moved from manufacturing engineering to process engineering to deepen my technical skill set, and later into tooling design to gain experience with a different product and system-level perspective.

As I move into a mid-career phase, I've realized that while I enjoy being hands-on technically, I'm also very interested in the bigger picture—how programs scale, how engineering decisions tie into business outcomes, and how organizations win and deliver work.

That's part of what attracts me to opportunities like ALTEN, where there's exposure not only to technical challenges but also to how engineering supports broader business goals."

Proudest Accomplishments

“I’ve found that I take the most pride in my work when I can clearly see its real-world impact.

Right now, working in commercial aviation has really reinforced that. It’s made me a bit of a plane nerd, but more importantly, I feel a strong sense of pride when I step onto an aircraft or travel across the world.

It puts my day-to-day work into perspective and reminds me that what I do contributes to systems that people rely on every day. That connection to a larger purpose is something I find very motivating.”

Questions for Alten

I did want to ask a quick clarification question ahead of time, and this might be more for Bailey but I initially applied for a Sustaining Mechanical Engineering role but after my screening I noticed the role is for a Level II position and based on my background and recent roles as a senior engineer, I wanted to confirm whether the Mechanical Engineer II level is the right fit for me or if there may be flexibility in leveling depending on experience.

What are some of the company's biggest upcoming milestones in the next 5 years?

"How would you describe the team's approach to work-life balance, especially during peak program periods?"

What does growth look like at Alten? (Individual Contributor to Management?)

How does Alten balance the standardization of different industries?

Example: Is there a challenge of having different Regulatory agencies like FDA for LifeSciences or FAA for Aerospace or __ for automotive?

What really attracted me to ALTEN is the diversity of projects and the emphasis on bringing together engineers with different backgrounds and perspectives.

From the research I've done, it seems that ALTEN values approaching problems from multiple angles, and that really resonates with me because I feel like I've learned so much from wearing different hats and being in different shoes throughout my career — whether it's working across design, manufacturing, materials, and suppliers — I

believe that seeing problems from different vantage points adds so much value and it’s reassuring to see a company culture that actively values that kind of breadth and diverse thinking.

Alten Research

Context:

My Role:

Key Decision:

Results:

Lessons Learned:

BL0 Project

Context:
One project that my team is going through right now is a design change for the joining of the Horizontal Stabilizer. There was an airplane configuration update where some parts were experiencing corrosion issues

My Role:

Key Decision:

Results:

Lessons Learned:

Laser Weld Capital Project

Context:

Our site had two laser welding machines: One was a modern IPG Welder bought in 2015. And the other was built in the 1990's basically on life support and it was getting more issues with calibration and parts were becoming obsolete. The machine was a risk to the factory if it were to go down, it would create a single point failure.

My Role:

My role as the Process Engineer for Laser Welding was to essentially spearhead the decision to purchase a Laser Welding machine for the site

This included providing justification for a new machine, working with the supplier for a new machine, coordinating with factory and facilities to schedule an installation plan, notifying the affected programs and creating documentation, presenting to PCCB,

Key decision:

We decided to go with the same company as our existing welder

Result:

It took a few months of rigorous work but we eventually got the new laser welder in. There was a minor hiccup about

Lesson:

What I thought was going to be a relatively simple process of copy and paste turned to be one of the most extensive projects mainly because it was a capital equipment requires more scrutiny and justification for purchase & half a million-dollar equipment

That's it

PCU Safety Project

Context:
Safety issue with PCU Installation // Finger jammed // Rudder to Vertical Fin Process
My Role
Design new tooling to make it easier for mechanics to install rudder to prevent safety hazard
Key decision
Different avenues to solving a problem
Result
Small issue with lifting / negotiating with Ergonomics [Polish]
Lesson
Different avenues to solving a problem
Done is better than perfect
That's it

Goldwing Program

Context:
When I was a Process Engineer at Raytheon, there was a program that needed more structure and organization Because there was another process engineer that wasn't really providing updates and the data would be really disorganized and I think he came from a program that didn't necessarily have the same culture of a space program so

My Role
Since I had previous experience on Space Programs as a Manufacturing Engineer so I was familiar with the expectations of what program management wanted in terms of the cadence and urgency and visibility. My manager moved me to this program to give more spark and transparency

Key decision
Worked with him extensively, got to know him on a personal level and understand him more. A lot of people viewed him negatively but I just gave him structure to showcase his work (Penn & Teller analogy)

Result
We scheduled daily tag up meetings,
I worked with him to create a template on his data to present better.
I took on the communication aspect so he could be more technical – I was almost a translator for him
Program was solved and delivered and customer was happy

Lesson
Working with process engineer that was brilliant but just lacked some soft skills and social skills

That's it